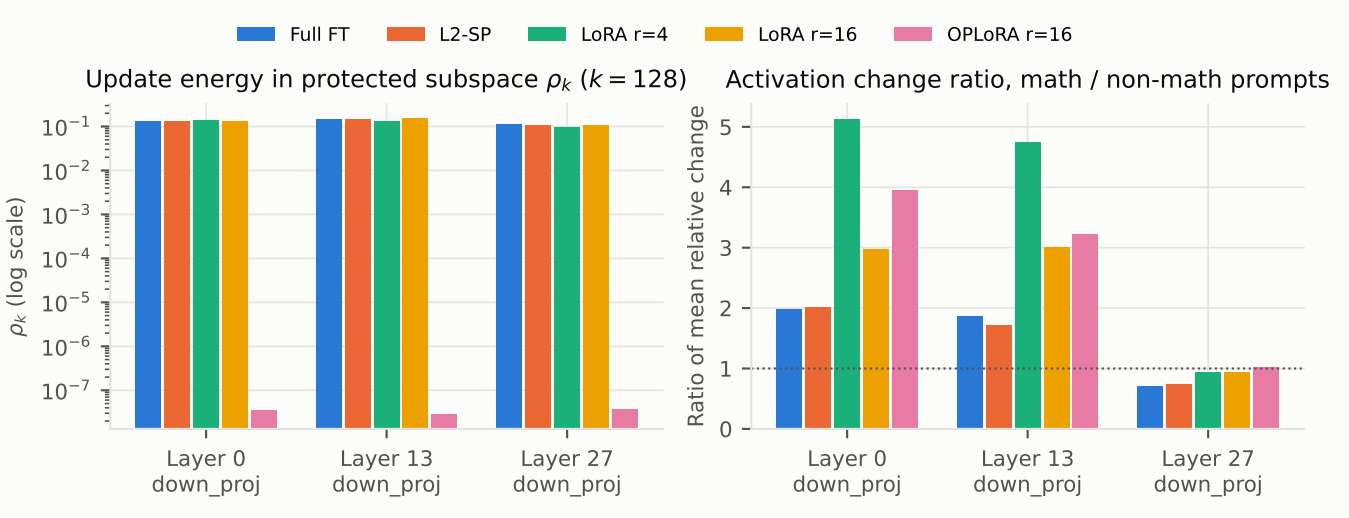
Full FT
L2-SP
LoRA r=4
LoRA r=16
OPLoRA r=16
Update energy in protected subspace ρ_k ($k = 128$)
ρ_k (log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
Layer 0 down_proj
Layer 13 down_proj
Layer 27 down_proj
Activation change ratio, math / non-math prompts
Ratio of mean relative change
0
1
2
3
4
5
Layer 0 down_proj
Layer 13 down_proj
Layer 27 down_proj